

IDMA (relations)

amir yehudayoff

pairs

two sets A, B

for $a \in A, b \in B$ the ordered pair (a, b)

recall the set $\{a, b\}$ is unordered

the set of pairs is (cartesian product)

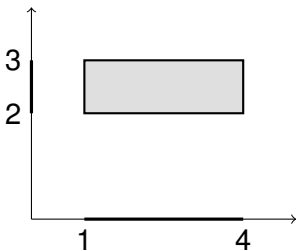
$$A \times B = \{(a, b) : a \in A, b \in B\}$$

the plane

the set $\mathbb{R}$ is the line

the set $\mathbb{R}^2$ is the plane

if $A, B \subset \mathbb{R}$ are intervals then $A \times B$ is a rectangle



more products

the product of $A_1, \dots, A_n$ is

$$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \dots, a_n) : a_i \in A_i \forall i \in [n]\}$$

with size

$$|A_1 \times A_2 \times \cdots \times A_n| = \prod_{i=1}^n |A_i|$$

relations

parent

child

know

bigger than

relations

two sets A, B

definition a relation is $R \subseteq A \times B$

for example $A = B$ are “people” and $(a, b) \in R$ if “ a parent of b ”

notation

a relation is $R \subseteq A \times B$

$$(a, b) \in R \iff aRb \iff R(a, b) = 1$$

if $R \subseteq A \times A$ then R is a relation on A

representations

a relation is $R \subseteq A \times B$

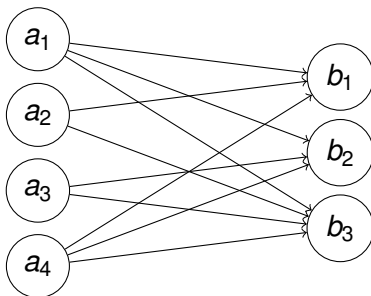
a boolean matrix

$$M_{a,b} = 1 \iff (a, b) \in R$$

a (bipartite) graph $G = (A \cup B, E)$

$$(a, b) \in E \iff (a, b) \in R$$

example



relative sets

a relation is $R \subseteq A \times B$

for $a \in A$ the relative set of A is

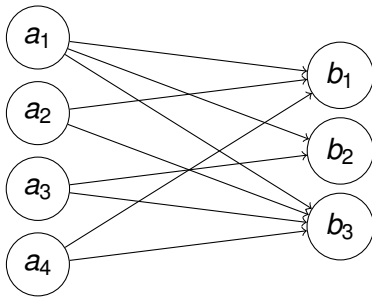
$$R(a) = \{b \in B : (a, b) \in R\} \subseteq B$$

e.g. the children of a

the relative set of $A' \subseteq A$

$$R(A') = \bigcup_{a \in A'} R(a)$$

example



$$R(a_2) = \{b_1, b_3\} \quad \text{and} \quad R(\{a_1, a_2\}) = B$$

relative sets

exercise

relation $R \subseteq A \times B$ and $A_1, A_2 \subseteq A$

if $A_1 \subseteq A_2$ then $R(A_1) \subseteq R(A_2)$

$$R(A_1 \cup A_2) = R(A_1) \cup R(A_2)$$

$$R(A_1 \cap A_2) \subseteq R(A_1) \cap R(A_2) \text{ (not always equal)}$$

types of relations

parent

know

greater than

what is special about each?

types of relations

$R \subseteq A \times A$ is a relation $a, b \in A$

symmetric $(a, b) \in R \Rightarrow (b, a) \in R$

“know”

asymmetric $(a, b) \in R \Rightarrow (b, a) \notin R$

“parent, greater than”

antisymmetric $(a, b) \in R \wedge (b, a) \in R \Rightarrow a = b$

“at least, divisible”

more types of relations

$R \subseteq A \times A$ is a relation $a, b, c \in A$

reflexive $(a, a) \in R$ for all $a \in A$

“know, at least”

irreflexive $(a, a) \notin R$ for all $a \in A$

“parent, greater than”

transitive $(a, b) \in R \wedge (b, c) \in R \Rightarrow (a, c) \in R$

“at least, greater than”

types of relations: matrices

$R \subseteq A \times A$ is a relation $\iff$ matrix M

reflexive ones on diagonal of M

irreflexive zeros on diagonal of M

symmetric $M = M^T$

transitive: matrices

$R \subseteq A \times A$ is a relation $\iff$ matrix M

define R^2 via the matrix $M \odot M$

$$(a, b) \in R^2 \iff \exists c \in A (a, c), (c, b) \in R$$

transitive: matrices

$R \subseteq A \times A$ is a relation $\iff$ matrix M

define R^2 via the matrix $M \odot M$

$$(a, b) \in R^2 \iff \exists c \in A (a, c), (c, b) \in R$$

claim R is transitive iff $R^2 \subseteq R$

transitive: matrices

$R \subseteq A \times A$ is a relation $\iff$ matrix M

define R^2 via the matrix $M \odot M$

$$(a, b) \in R^2 \iff \exists c \in A (a, c), (c, b) \in R$$

claim R is transitive iff $R^2 \subseteq R$

proof sketch

if R is transitive and $(a, b) \in R^2$ then $(a, b) \in R$ via $\exists c$

if $R^2 \subseteq R$ and $(a, c), (c, b) \in R$ then $(a, b) \in R^2 \subseteq R$

equivalences

one of the most important concepts is “equality”

we ignore the “unimportant stuff”

examples

$$\frac{1}{2} = \frac{2}{4}$$

$$0 = 12 \pmod{2}$$

$$0.9999999 \dots = 1$$

equivalences

definition

a relation R on A is an equivalence relation if it is

- reflexive (aRa)
- symmetric ($aRb \iff bRa$)
- transitive ($aRb, bRc \Rightarrow aRc$)

equivalence: rationals

for

$$\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$$

define “equality” via

$$\frac{a}{b} R \frac{c}{d} \iff ad = bc$$

claim this is an equivalence relation

equivalence: modular arithmetic

for integer $n > 0$ define a relation on $\mathbb{Z}$ via

$$aRb \iff \exists c \in \mathbb{Z} \ a = b + cn$$

for example for $n = 3$ the following pairwise satisfy R

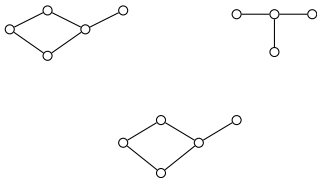
$$\dots, -3, 0, 3, 6, 9, \dots$$

claim this is an equivalence relation

equivalence: connectivity

$G = (V, E)$ is an undirected graph and define R on V via

$$uRv \iff \text{there is a path from } u \text{ to } v$$



claim this is an equivalence relation

equivalence classes

R an equivalence relationship on A

definition the equivalence class of $a \in A$ is

$$R(a) = \{b \in A : (a, b) \in R\}$$

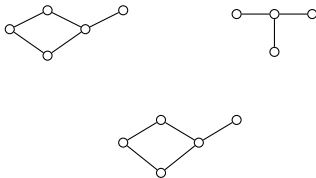
in words, all things equivalent to a

equivalence classes: examples

“modulo 4” over $\mathbb{Z}$

$$R(1) = \dots, -3, 1, 5, 9, 13, \dots$$

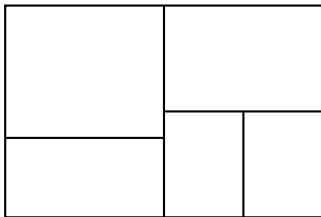
“connected” in graph: connected components



partitions

definition a partition of A is a collection $P_1, \dots, P_m \subseteq A$ of pairwise disjoint sets¹ such that

$$A = \bigcup_{i=1}^m P_i$$



¹for all $i \neq j$ we have $P_i \cap P_j = \emptyset$

partitions are ER

claim (exercise)

1. if R is an equivalence relation on A then the set²
 $\{R(a) : a \in A\}$ is a partition of A

²there are no repetitions in a set

partitions are ER

claim (exercise)

1. if R is an equivalence relation on A then the set²
 $\{R(a) : a \in A\}$ is a partition of A
2. if $P_1, \dots, P_m$ is a partition of A then R defined by

$$aRb \iff \exists i \in [m] \ a, b \in P_i$$

is an equivalence relation

²there are no repetitions in a set

partitions are ER

claim (exercise)

1. if R is an equivalence relation on A then the set²
 $\{R(a) : a \in A\}$ is a partition of A
2. if $P_1, \dots, P_m$ is a partition of A then R defined by

$$aRb \iff \exists i \in [m] \ a, b \in P_i$$

is an equivalence relation

comment to prove need to use definitions and verify definitions

²there are no repetitions in a set

representatives

R is an equivalence relation over A

from each equivalent class we can choose a representative

example for “equality” of $\mathbb{Q}$ choose $\frac{1}{2}$ from

$$\left\{ \frac{1}{2}, \frac{2}{4}, \frac{3}{6}, \frac{4}{8}, \frac{5}{10}, \frac{-6}{-12}, \dots \right\}$$

operations on relations

let $R \subseteq A \times B$ be a relation

the complement of R is

$$\overline{R} = \{(a, b) \in A \times B : (a, b) \notin R\}$$

the inverse of R is

$$R^{-1} = \{(b, a) \in B \times A : (a, b) \in R\}$$

what is the meaning in matrix language?

posets

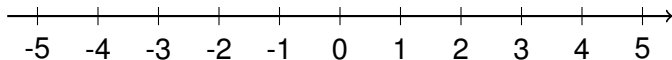
some things are incomparable (pros and cons)

definition a relation R on A is a partial order if it is reflexive, antisymmetric and transitive

$$a \leq a, a \leq b \Rightarrow b \not\leq a, a \leq b, b \leq c \Rightarrow a \leq c$$

posets: examples

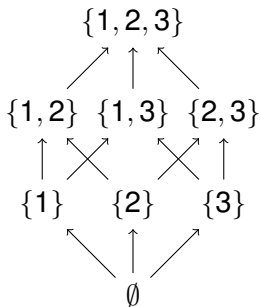
the integers with the usual $\leq$



posets: examples

the family A of all subsets of S ordered by inclusion

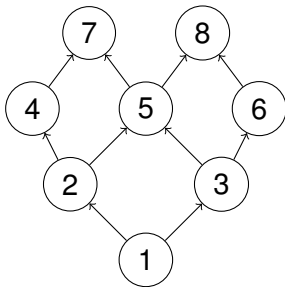
$$a \leq b \iff a \subseteq b$$



posets: examples

the vertices of a DAG³

$a \leq b \iff$ there is a path $a \rightarrow b$



³directed acyclic graph

Hasse diagrams

we can draw a poset $(A, \leq)$

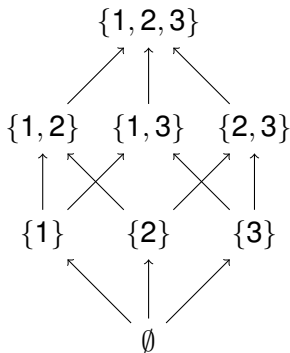
the vertices are $a \in A$

the edges are the “minimal” $a \rightarrow b$ with $a \leq b$

—remove self-loops

—remove edges implied by transitivity

—direct edges “upward”



rooted directed trees

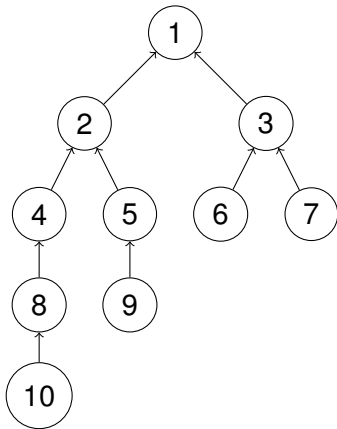
important type of DAGs

rooted directed tree is (V, E) with $E \subseteq V^2$

special vertex $r \in V$ so that for every $v \in V$ there is path $r \rightarrow v$

for every $v \neq u$ in V there is at most one path $v \rightarrow u$

rooted directed trees: example



rooted directed trees

rooted directed trees appear in many applications

data structures—efficiently maintain lists of numbers

computation—represent formulas

decision trees—represent “options”

terminology for rooted directed tree

root

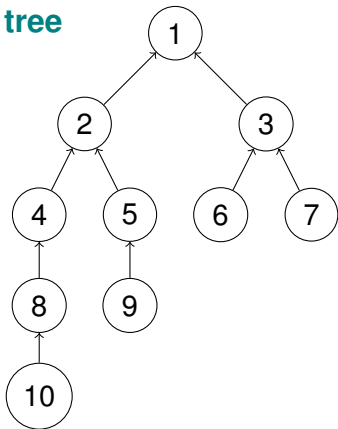
leaves: no out-going edges

parent and child: $(v, u) \in E$

siblings: $(v, u), (v, w) \in E$

subtree rooted at $v \in V$

height or depth: length of longest directed path



extremes

“best”, “worst”, ...

extremes

“best”, “worst”, ...

for a poset $(A, \leq)$

maximum $a \in A$ such that $b \leq a$ for all $b \in A$

minimum $a \in A$ such that $a \leq b$ for all $b \in A$

extremes

“best”, “worst”, ...

for a poset $(A, \leq)$

maximum $a \in A$ such that $b \leq a$ for all $b \in A$

minimum $a \in A$ such that $a \leq b$ for all $b \in A$

maximal $a \in A$ such that there is no $b \in A$ so that $a < b$

minimal $a \in A$ such that there is no $b \in A$ so that $b < a$

extremes

“best”, “worst”, ...

for a poset $(A, \leq)$

maximum $a \in A$ such that $b \leq a$ for all $b \in A$

minimum $a \in A$ such that $a \leq b$ for all $b \in A$

maximal $a \in A$ such that there is no $b \in A$ so that $a < b$

minimal $a \in A$ such that there is no $b \in A$ so that $b < a$

e.g. $[n]$, $\mathbb{Z}$, $\mathbb{N}$, $\mathcal{P}(1, 2, 3)$

are there extremes?

claim let A be a finite set and $\leq$ a partial order

there are always maximal elements

there are always minimal elements

there is at most one maximum element

there is at most one minimum element

closure

what if a relation is not transitive (for example)

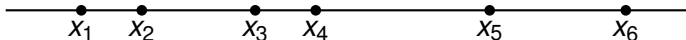
can “make” R transitive

internally: keep adding things until transitive

externally: smallest transitive relation containing

closure: example

goal: the smallest interval containing points $X = \{x_1, \dots, x_6\}$



external: $\bigcap I$ where I is interval containing X

internal: keep adding new points in between...

intersections

R, Q are relations on A

$R \cap Q$ is a relation on A

intersections

R, Q are relations on A

$R \cap Q$ is a relation on A

claim

if R, Q are reflexive then so is $R \cap Q$

if R, Q are symmetric then so is $R \cap Q$

if R, Q are transitive then so is $R \cap Q$

reflexive closure

the reflexive closure R^{ref} of R is the minimum (with respect to inclusion) reflexive relation that contains R

claim for all relations R on A

$$R^{ref} = \bigcap_S S = R \cup \{(a, a) : a \in A\}$$

intersection is over all reflexive relations S on A that contain R

symmetric closure

the symmetric closure R^{sym} of R is the minimum (with respect to inclusion) symmetric relation that contains R

claim for all relations R on A

$$R^{sym} = \bigcap_S S = R \cup R^{-1}$$

intersection is over all symmetric relations S on A that contain R

transitive closure

the transitive closure R^{sim} of R is the minimum (with respect to inclusion) transitive relation that contains R

$$R^{tran} = \bigcap_S S = ?$$

intersection is over all transitive relations S on A that contain R

transitive closure

$$R^{tran} = ?$$

recall

R is described by boolean matrix M

R^2 is defined by $M^{\odot 2}$

transitive closure

$$R^{tran} = ?$$

recall

R is described by boolean matrix M

R^2 is defined by $M^{\odot 2}$

for all integer $d > 0$ define R^d via $M^{\odot d}$

transitive closure

$$R^{tran} = ?$$

recall

R is described by boolean matrix M

R^2 is defined by $M^{\odot 2}$

for all integer $d > 0$ define R^d via $M^{\odot d}$

theorem

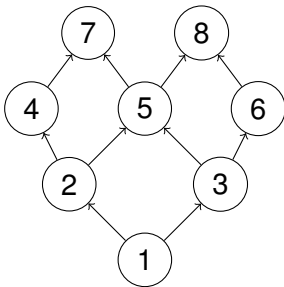
$$R^{tran} = R^{\infty} := \bigcup_d R^d$$

transitive closure: examples

transitive closure of edges

the vertices of a DAG

$$a \leq b \iff \text{there is a path } a \rightarrow b$$



linear order

a linear order⁴ on A is partial order $\leq$ on A so that every two elements are comparable

that is, for all $a, b \in A$ either $a \leq b$ or $b \leq a$

examples

$(\mathbb{Z}, \leq)$ is linear

$(\mathcal{P}(S), \subseteq)$ is not linear

⁴or a full order

extensions

claim (exercise)

for every partial order R on a finite set A there is linear order $\leq$ that extends R

$$aRb \Rightarrow a \leq b$$

usually, there are many

summary

relations

types of relations

operations on relations

equivalences

partial and linear orders